Date :

sum = 6+4 = 10

$\boxed{sum=10}$

i = 4+1 = 5

5 <= 5 ✓

sum = 10+5 = 15

$\boxed{sum=15}$

i = 5+1 = 6

6 <= 5 X

out of loop

print sum 15

Q. write a program to find the factorial of a number.

29/1/26

=>

5×4×3×2×1

```
#include <stdio.h>
void main()
{
        n×(n-1)×(n-2)×(n-3)×(n-4)
   int i, n, fact;
   printf("Enter n");
   scanf("%d", &n);
   fact = 1;
   for (i=1; i<=n; i=i+1)
   {
        fact = fact*i;
   }
printf("%d", fact);
}
```

— write the operation to read and display a 2x2 matrix

2- Dimensional array

* write a program to read and print a two dimensional array.

```
#include<stdio.h>

int main()
 {
   int a[100][100] ,i,j,m,n;
   printf("rowsize and columnsize");
   scanf("%d%d", &m,&n);
   printf("Enter matrix elements");

   for(i=0;i<m; i++)
   {
      for(j=0;j<n;j++)
      {
         scanf("%d", &a[i][j]);
      }
   }
 for(i=0;i<m;i++)
 {
    for(j=0;j<n;j++)
    {
     printf("%d \t", a[i][j]);
    }
```

```c
printf("\n");
}
    return 0;
}
```

10/10

Ø Write the operation to read and display a 2x2 matrix

```c
#include<stdio.h>

int main() {
    int m,n,i,j, a[100][100];

    printf("Enter the row number and column number:");
    scanf("%d%d", &m, &n);

    printf("Enter the matrix ~~numbers~~ elements: ");
    for(i=0; i<m; i++)
    {
        for(j=0;j<n;j++)
        {
            scanf("%d", &a[i][j]);
        }
    }

    for(i=0; i<m; i++)
    {
        for(j=0;j<n;j++)
```

Date

① – Write a program using two different categories of function to read a 3 digit number and find the reverse of that number using reverse function and find sum of digit of that number using sum function.

OP verified ⑤ – Write a program to perform bubble sort using fun with argument but with no return type.

– Write a program to find the difference between two matrices using function with argument and no return type.

verified ① ⑤

```
#include <stdio.h>

void sum(int);
int reverse(int);
int main()
{
 int a,r;

 printf("Enter the digit:");
 scanf("%d",&a);

 r= reverse(a);
 printf("The reverse of the number is %d\n",r);
```

i=i+1;

}

printf("The sum of first %d natural numbers is %d", n, sum);

}

o/p: Enter the value of n 5

The sum of first 5 natural numbers is 15

$\boxed{i=1}$ $\boxed{sum=0}$ $\boxed{n=5}$

1<=5 ✓

sum = 0+1 = 1

i = 1+1 = 2

$\boxed{i=2}$ $\boxed{sum=1}$ $\boxed{n=5}$

2<=5 ✓

sum = 1+2 = 3

i = 2+1 = 3

$\boxed{i=3}$ $\boxed{sum=3}$ $\boxed{n=5}$

3<=5 ✓

sum = 3+3 = 6

i = 3+1 = 4

$\boxed{i=4}$ $\boxed{sum=6}$ $\boxed{n=5}$

4<=5 ✓

sum = 6+4 = 10

i = 4+1 = 5

$\boxed{i=5}$ $\boxed{sum=10}$ $\boxed{n=5}$

$5 <= 5$ ✓

$sum = 10+5 = 15$

$i = 5+1 = 6$

$\boxed{i=6}$ $\boxed{sum=15}$ $\boxed{n=5}$

$6 <= 5$ X

sum= 15 print.

② for loop

$\boxed{i=1}$ $\boxed{n=5}$ $\boxed{sum=0}$

$1 <= 5$ ✓

~~[illegible]~~ $sum = 0+1 = 1$ $i = 1+1 = 2$

$\boxed{i=2}$ $\boxed{n=5}$ $\boxed{sum=1}$

$2 <= 5$ ✓

$sum = 1+2 = 3$ $i = 2+1 = 3$

$\boxed{i=3}$ $\boxed{n=5}$ $\boxed{sum=3}$

$3 <= 5$ ✓

$sum = 3+3 = 6$ $i = 3+1 = 4$

$\boxed{i=4}$ $\boxed{n=5}$ $\boxed{sum=6}$

$4 <= 5$ ✓

$sum = 6+4 = 10$ $i = 4+1 = 5$

$\boxed{i=5}$ $\boxed{n=5}$ $\boxed{sum=10}$

$5 <= 5$ ✓ $sum = 10+5 = 15$ $i = 5+1 = 6$

$\boxed{i=6}$ $\boxed{n=5}$ $\boxed{sum=15}$

6<=5 X

out of loop

print sum = 15.

```
#include<stdio.h>
void main()
{
  int i,sum,n;
  sum=0;
  printf("Enter the value of n");
  scanf("%d",&n);
  for (i=1; i<=n; i++)
   {
     sum= sum+i;
   }
printf("The sum of first %d natural numbers is %d",n,sum);
}
```

(3) <u>do... while</u>

```
#include<stdio.h>
void main()
{
int i,sum,n;
i=1;
```

```
sum=0;
printf("Enter the value of n");
scanf("%d", &n);
do
  {
    sum= sum+i;
    i=i+1;
  }
while (i<=n);
printf("The sum of first %d natural numbers is %d", n, sum);
}
```

$\boxed{i=1}$ $\boxed{n=5}$ $\boxed{sum=0}$

$sum=0+1=1$

$\boxed{sum=1}$

$i=1+1=2$

$2<=5$ ✓

$sum=1+2=3$

$\boxed{sum=3}$

$i=2+1=3$

$3<=5$ ✓

$sum=3+3=6$

$\boxed{sum=6}$

$i=3+1=4$

$4<=5$

1. Write a program to swap two numbers using function without return type and without arguments.
2. Without return type and with arguments.

- Actual arguments - fn call arguments.
- Formal arguments - fn definition. arguments.

3. ~~pass by reference / call by reference.~~

Recursion

A function calls itself is known as recursion.

Copying and comparing structure variables

Two variables of the same structure type can be copied as the same way as ordinary variables. If book1, and book2 are the two variables of the struct book, then the following are valid.

book1 = book2;

However the statement such as book1 == book2 is not permittable. C doesn't permit any logical operations on structure variables. If we ~~don't~~ want to compare them, we may do do by comparing members individualy.

Array of structures

→ Write a program to declare a structure with structure name marks. Structure members an int value total mark. and an int array of sub, which can store marks of 3 subjects. Structure variable is an array of structure called student that can store the details of 3 students.

Q. Write a program to calculate the total marks of all subjects of each students. Also calculate the total marks gained by each student for each subject.